



Updated DFD, Activity, ER Diagram

Course Title : Software Project Design and Development

Course Code : CSE 416

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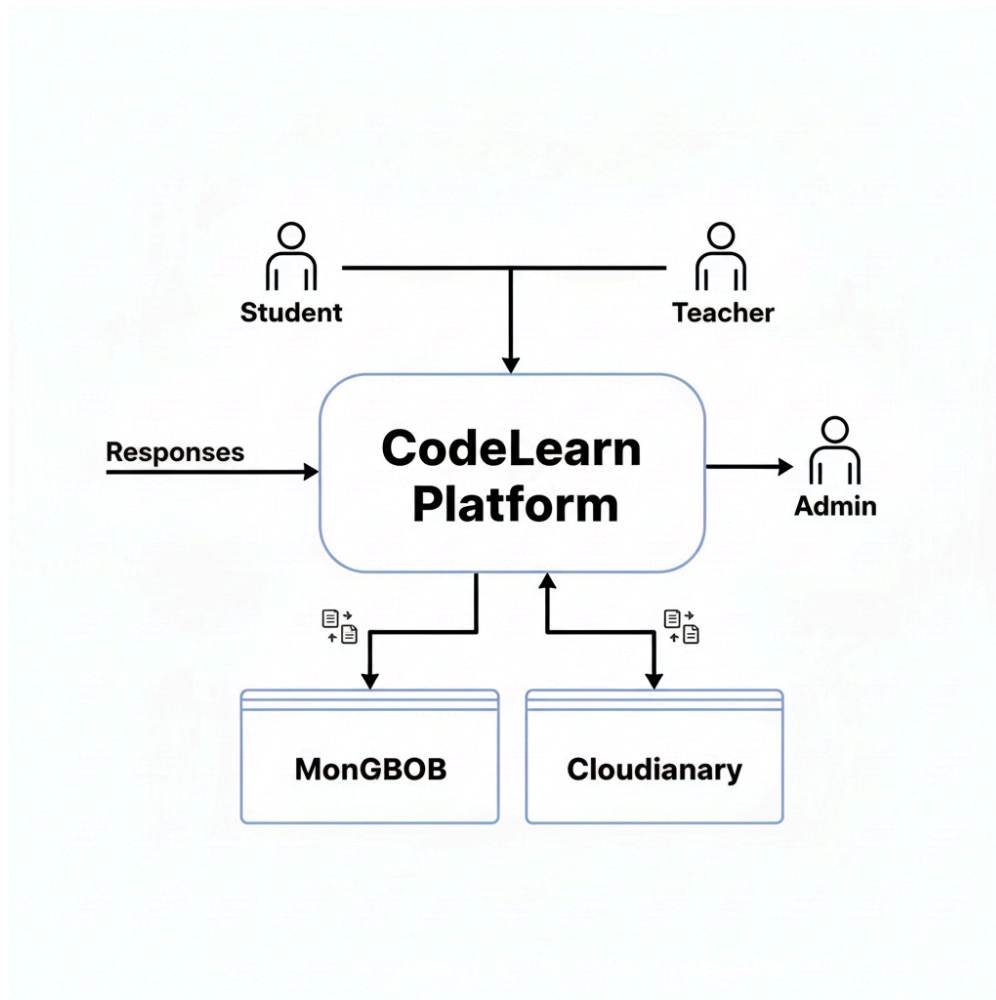
Section : 7B

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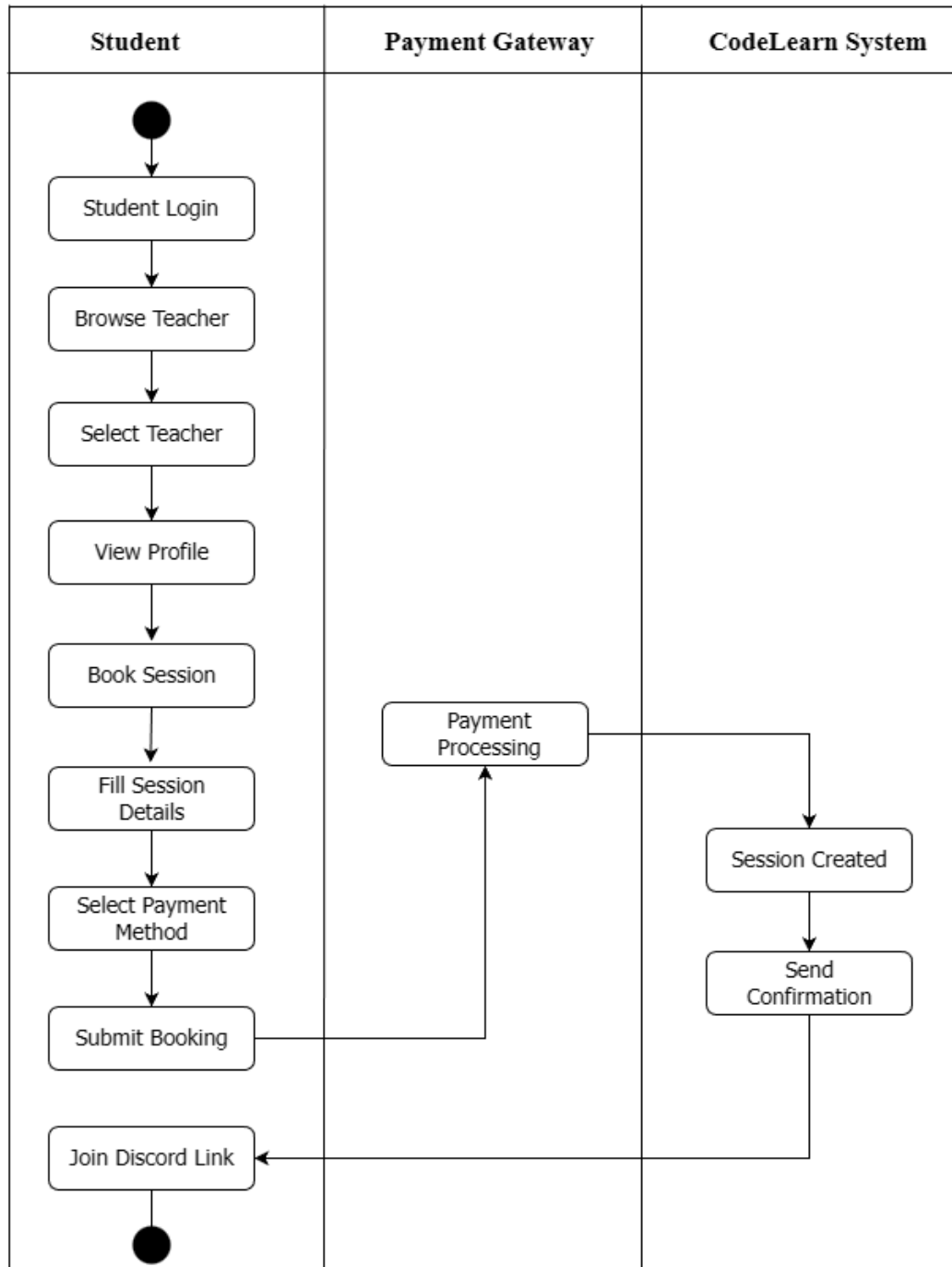
DFD (Data Flow Diagram)



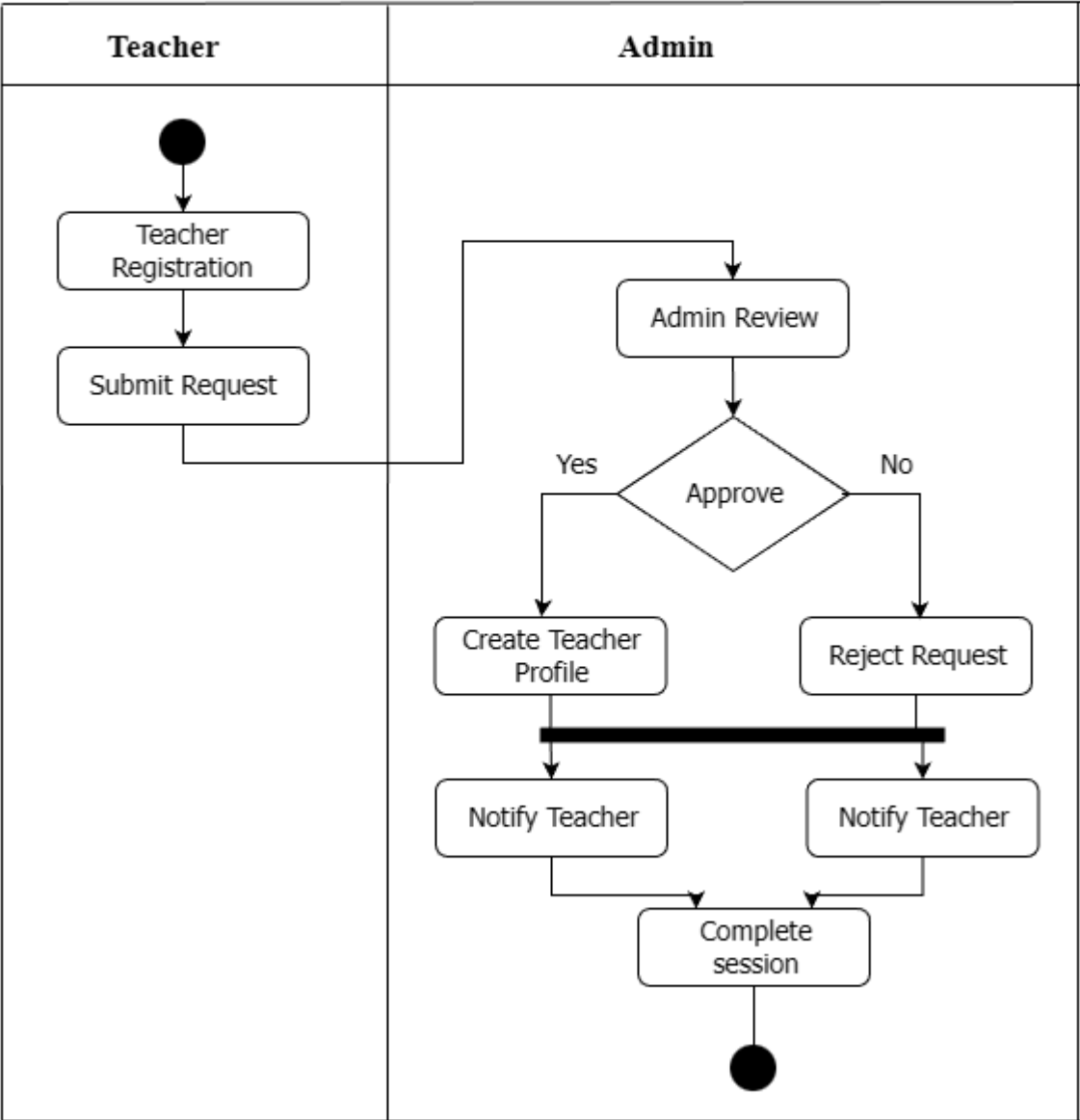
The diagram shows the **CodeLearn Platform's three-tier architecture**. The **Platform** is the central processing unit that handles all logic (**matching, booking, real-time sessions**). It interacts with three main **Actors (Student, Teacher, Admin)** and connects to **two storage** systems: **MongoDB** (for application data like profiles and session records) and **Cloudinary** (for media assets like profile pictures and documents).

Activity Diagram for (Student, Teacher, Messages)

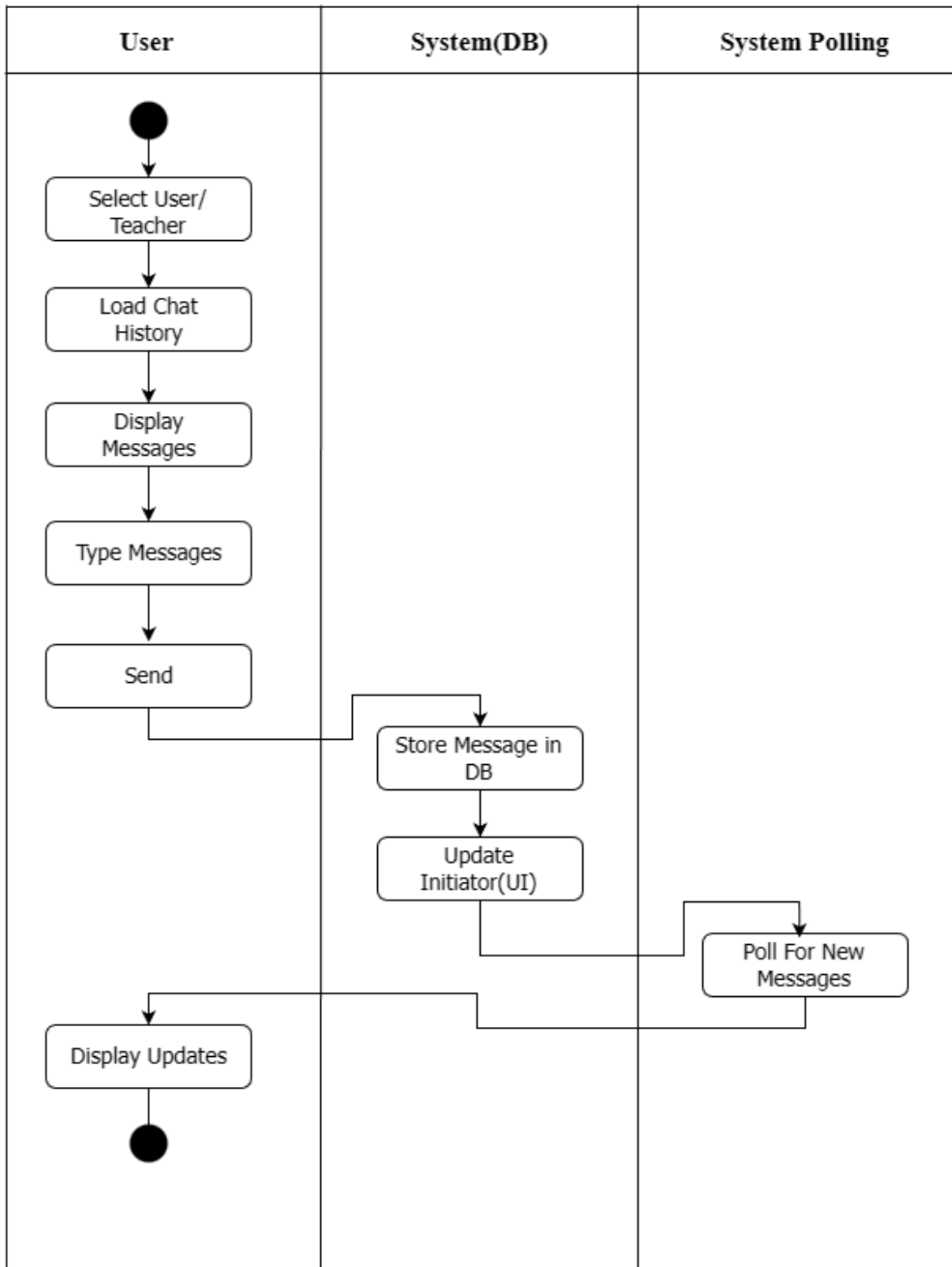
CODELEARN : USER ACTIVITY DIAGRAM



CODELEARN : TEACHER ACTIVITY DIAGRAM

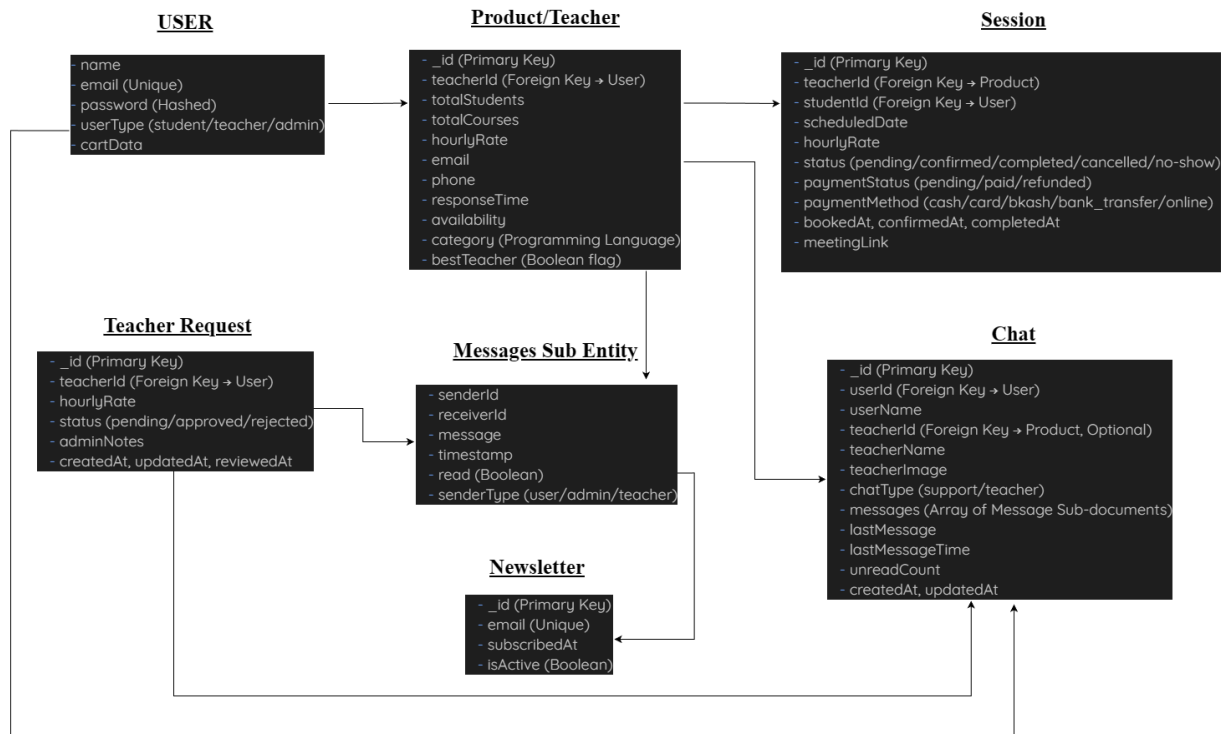


CODELEARN : CHAT ACTIVITY DIAGRAM



Entity Relationship Diagram

CODELEARN: ENTITY RELATION DIAGRAM



This Entity Relation Diagram (ERD) for CodeLearn models the platform's database structure, defining how key pieces of information relate to one another.

The core entities are:

USER: The base account for all roles (Student, Teacher, Admin).

Product/Teacher: Details specific to a teacher, including their expertise, hourly rate, and total courses, linked to the base USER entity.

Session: Records of booked tutoring events, linking a specific teacherId and studentId, along with status, payment details, and the meeting link.

Chat: Manages communication channels (direct or support) between users.

Teacher Request: A separate entity to handle the application/vetting process for new teachers before they become an active Product/Teacher.